

Summer Internship Mid Review

SecureQRouter: A Q-Learning Based Intrusion-Resilient and Privacy-Aware Routing Protocol for IoT Mesh Networks

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Introduction

- IoT mesh networks (smart agriculture, industrial monitoring, disaster response) rely on multi-hop routing where nodes are energy-constrained, topology changes frequently, and no central controller is always reachable.
- Q-Learning-based geographic routing lets nodes learn efficient next-hop decisions from local delay/energy feedback rather than a fixed routing table – the base paper this internship builds on.
- The base paper's Q-router assumes a *fully cooperative* network: it has no mechanism to authenticate neighbours or detect a manipulated reward signal.
- This leaves it vulnerable to Blackhole, Sybil, Wormhole, and Byzantine behaviour, and to routing decisions built on self-reported, spoofable node metrics.
- **SecureQRouter** extends the same Q-Learning router with a non-invasive, locally-verifiable trust term – the underlying algorithm is preserved, not replaced.

Problem Statement

- A Blackhole node can advertise favourable Q-values and then silently drop every packet routed through it – the base router has no delivery-verification step to catch this.
- Sybil, Wormhole, and Byzantine nodes each attack a different part of the routing pipeline: the neighbour set, the topology model, and the Q-Learning convergence assumption respectively.
- Existing trust-aware schemes either compute trust centrally (single point of failure) or rely on metrics the neighbour itself reports – and can therefore misreport.
- **Research question:** Can a locally-verifiable trust term be added to an existing Q-Learning router's reward so that routing degrades *gracefully*, not catastrophically, under Blackhole/Sybil/Wormhole/Byzantine nodes – without a centralised controller and without replacing the underlying learning algorithm?

Literature Survey

Table 1: Summary of Literature Survey (representative subset)

Sl. No	Author(s) & Year	Focus	Approach & Key Finding	Gap Addressed by Secure-QRouter
1	Kamgueu et al., 2021	Q-RPL: Q-Learning in RPL parent selection	+15% PDR over standard RPL under mobile topologies	Fixed reward structure; no security at all
2	Malik et al., 2022	DQNSec: security-aware DRL routing (OpploT)	Mitigates sinkhole, hello-flood, DDoS via a security-aware reward	Limited to three attacks; deep-network overhead
3	Coronado et al., 2022	R3-IoT: reliability-aware RL routing	Folds node reliability (temperature, CPU load) into routing	Reliability metrics are self-reported – trivially spoofable
4	Ranjith et al., 2025	Trust-aware DRL for SD-WSN security	COA-based router + HEC signcryption; joint energy/delay/trust/security	Depends on an SDN controller – single point of failure
5	Sharma et al., 2023	QSec-RPL: RL attack detection in RPL	> 93% accuracy detecting version-number attacks	Specialised to one attack type; does not generalise

Motivation

- Across the fifteen studies reviewed, security is a first-class design goal in only six – the rest, including the base paper, assume a fully cooperative network.
- Where security *is* addressed, it is narrow (one or three attack types) or reintroduces a centralised controller/self-reported metric – exactly the two things a distributed mesh router is meant to avoid.
- This motivates a router that is (i) fully distributed, (ii) verifies rather than trusts locally-reported behaviour, and (iii) is evaluated against a *broad* set of attacks rather than one.
- A reproducible, multi-seed, statistically-tested evaluation is needed to confirm any observed resilience is real and not an artefact of one random topology or one random seed.

Objectives

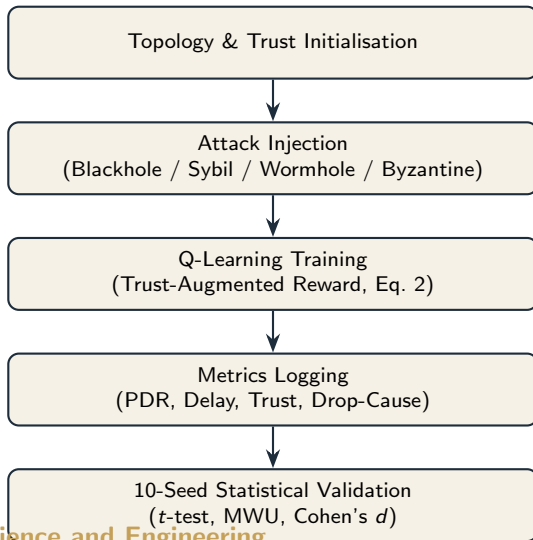
- Extend the base paper's Q-Learning reward with a trust term computed from locally-observable, non-self-reported behaviour (delivery acknowledgements, forwarding rate).
- Implement an attack-injection framework covering Blackhole, Sybil, Wormhole, and Byzantine node behaviour inside the same simulator used for baseline evaluation.
- Evaluate SecureQRouter against the trust-unaware baseline across three network topologies and an infection-intensity sweep (0–60%).
- Establish, using paired non-parametric tests (Mann-Whitney U) and paired t -tests with Cohen's d , whether any observed change is statistically significant rather than an artefact of a particular seed.

Proposed Methodology

- Three topologies are evaluated: a dense 100-node random deployment, a sparser 38-node mesh, and a 21-node tree with the least path redundancy – one fixed sink per topology.
- The routing engine keeps the base paper's Q-table and ε -greedy policy *unchanged*; only the reward computation is augmented with a trust term (non-invasive by design).
- An attack-injection layer marks a configurable fraction of nodes as Blackhole, Sybil, Wormhole, or Byzantine, evaluated by the same routing and logging code as honest nodes.
- The infection-intensity sweep varies malicious-node fraction from 0% to 60% in 10-point steps; a separate per-attack-type log runs ten independent seeds (42–51) at each intensity.

- Paired t -tests, Mann-Whitney U , and Cohen's d are computed between the 0%

Proposed Methodology



System Architecture

- Random 100-node deployment (20×20 km, 5.0 km range) – densely connected, multiple redundant paths to the sink.
- Sparser 38-node mesh (25×25 km, 4.5 km range) and a 21-node tree (50×50 km, 4.0 km range) – progressively less path redundancy, isolating topology effects from attack effects.
- Every node maintains a locally-computed trust score; the router consults only its own table and its own observations – no coordinator, no shared state between nodes.

Results & Discussion

Table 2: Routing Performance Across Infection Intensity

Infection (%)	PDR (%)	Delay (ms)	Connectivity (%)	Delivered	Avg. Trust
0	1.67	1.50	99	100	0.6109
10	1.25	1.33	90	150	0.6009
20	2.06	1.54	79	370	0.5823
30	4.25	2.09	62	1,020	0.5640
40	5.53	2.22	16	1,660	0.5327
50	5.00	2.12	17	1,800	0.4861
60	6.60	2.23	23	2,770	0.4443

- Average trust falls monotonically from 0.611 to 0.444 across the sweep – the strongest evidence that the trust mechanism responds correctly to adversarial presence.
- PDR stays low throughout (1.7% $\rightarrow$ 6.6%); the next slide shows *why*.

Results & Discussion

Table 3: Packet Drop-Cause Breakdown at 60% Infection (42,000 packets)

Outcome / Drop Cause	Packets	Share
Delivered	2,770	6.6%
Dropped – routing loop	38,510	91.7%
Dropped – no route available	280	0.7%
Dropped – interaction with infected node	250	0.6%

- **Honest finding:** at the highest infection level, injected attacks cause only 0.6% of drops – **91.7% of packet loss is a routing-loop issue, largely independent of infection level.**
- This explains why PDR is already low (1.67%) even at 0% infection, and means PDR alone should *not* be read as a direct measure of attack resilience.

Results & Discussion

Table 4: Mean Routing Metrics by Attack Type (10 seeds per configuration)

Attack	Intensity	PDR (%)	PDR SD	Delay (ms)	Detection (%)	FPR (%)
Blackhole	0	99.40	1.28	2.67	100.0	0.0
Blackhole	10	31.50	12.91	1.36	92.0	0.0
Blackhole	20	16.66	8.43	0.63	87.0	0.0
Byzantine	0–20	99.40	1.28	2.67	100.0	≤ 0.2
Sybil	0–10	99.40–99.72	≤ 1.28	2.66–2.68	100.0	0.0

- Blackhole PDR collapses steeply and seed-consistently (99.4% $\rightarrow$ 16.7%) – behaves exactly as the threat model predicts.
- **Byzantine's row is not a typo:** every metric is numerically identical across all tested intensities, per seed – a confirmed injection defect, not evidence of resilience.
- Sybil shows only a narrow, non-monotonic PDR band – a much weaker response than Blackhole, flagged for the same reason.

Results & Discussion

Table 5: Paired Significance Tests – Baseline (0%) vs. Highest Intensity Logged

Attack	Metric	Paired t	p (paired t)	Cohen's d	Sig. ($\alpha = 0.05$)
Blackhole	PDR (0% vs. 20%)	32.192	1.32×10^{-10}	-13.720	Yes
Blackhole	Connectivity (0% vs. 20%)	0.000	1.0000	0.000	No (expected)
Byzantine	All 4 metrics (0% vs. 20%)	0.000	1.0000	0.000	No – confirms bug
Sybil	Energy (0% vs. 10%)	-4.759	0.0010	0.287	Yes (isolated)
Wormhole	All 4 metrics (0% vs. 3%)	$\approx \pm 1.0$	> 0.34	small	No (underpowered)

- Blackhole: large, highly significant effect on PDR/delay/energy; correctly *no* effect on connectivity (a graph property, not attack-dependent).
- Byzantine's $t = 0$, $p = 1$ on every metric *confirms* – not just suggests – that its intensity parameter never reaches routing outcomes.
- Wormhole's narrower 0–3% range (vs. 0–20% for the others) leaves this comparison underpowered rather than conclusively negative.

Future Work

Planned Directions

- Investigate and fix the routing-loop bug identified in the drop-cause breakdown – currently the dominant limiter of PDR across the entire sweep.
- Audit and fix the Byzantine and Sybil attack-injection paths against the working Blackhole implementation.
- Extend the Wormhole log from its current 0–3% range to match the 0–20% range used for Blackhole/Byzantine.
- Repeat the infection-intensity sweep with multiple seeds per level to separate genuine topology effects from single-run variance.
- Capture the Q-Learning training log at a finer interval to plot a proper convergence curve; extend validation to physical IoT hardware.

Conclusion

Summary

The trust mechanism in SecureQRouter works as intended: average trust falls monotonically and significantly as infection intensity rises, and Blackhole shows a large, statistically confirmed decline in PDR, delay, and energy with no spurious effect on connectivity. This resilience finding does *not* yet extend to Byzantine, Sybil, or Wormhole – formal significance testing confirms a Byzantine injection defect (every test statistic exactly zero), an isolated and cautious Sybil effect, and an underpowered Wormhole comparison. Separately, 91.7% of packet loss at peak infection is a routing-loop artefact rather than attack-driven, so PDR alone should not be read as a measure of attack resilience until that bug is fixed. These are reported as open, verifiable engineering items rather than smoothed over, consistent with the statistically rigorous evaluation this internship set out to deliver.

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Thank You

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